Homework 1

For this homework you will create a github repo, set up github pages, clone the repo to your computer as an `R project`, create a `.qmd` file, and push those changes back to github to create a webpage! You'll submit the link to your github pages site (the one that renders as a nice website).

Sounds like a lot. We'll get through it! (We're going to follow the process outlined for 'render to docs' from here.)

Step 1

- Head to github and create a new repo.
  - Be sure to make the repo public and **do not** choose a `.gitignore` for this one

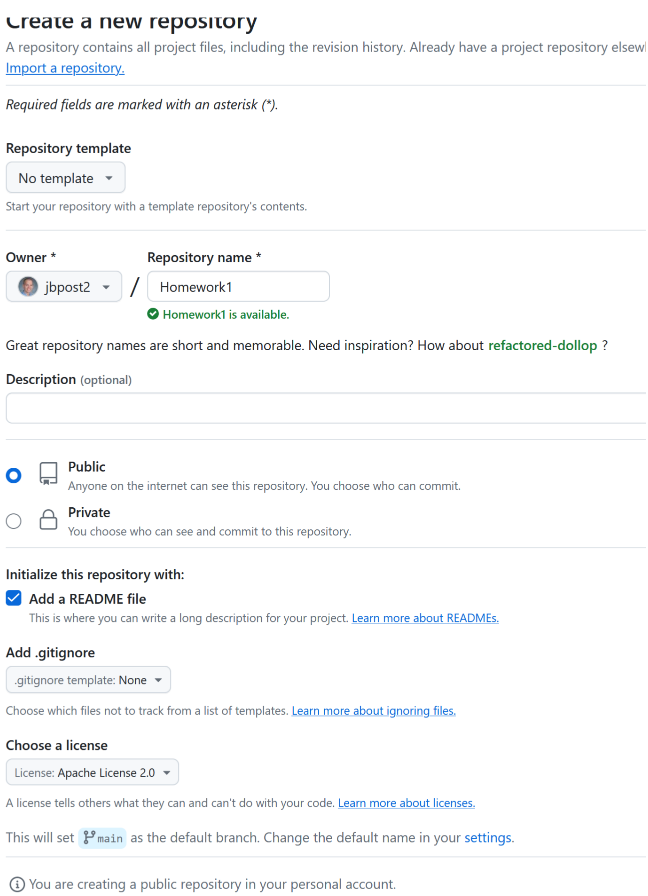

Figure 1: Github repo creation form.

Step 2

- Create a new `R project` from version control (as we did in the notes/videos) that clones this repository locally.
  - Recall there is a (green) '<> Code' button to copy the repo link.
  - A `.gitignore` file may be created in this process. That isn't a worry!

Step 3

- Within `RStudio`, now create a new `.qmd` document that outputs to `HTML`. Give this a title about data science. Save the file in the main repo folder on your computer.
- In this `quarto` document, we want to do two things (separate them with headers):
  - First, write text to answer the question:
    * What do you think being a data scientist is about?
    * What differences/similarities do you see between data scientists and statisticians?
    * How do you view yourself in relation to these two areas?
  - You can write in a conversational tone or more formally (however you want to represent yourself). **Use a markdown list at some point.** There is no word count or anything like that, just make sure you answer the prompts above to receive full credit.
  - Second, include a section with the following R code:

```
y <- density(iris$Petal.Width)
```

    * Create an `R` code chunk to determine the class, type, and structure of the object `y`
    * Create an `R` code chunk that uses the plot function on `y`. Hide the `R` code in the final document by setting `echo` to `FALSE`.
    * Include some markdown text between these code chunks explaining what you are doing with the `R` code.
  - You can render the document as needed to check things are looking good locally.

Step 4

In your repo folder on your computer, create a file called `_quarto.yml`. Open this file (perhaps in `RStudio` or a text editor) and place the following in the file (spacing is important - the spacing must match below exactly - two spaces prior to `type` and `output-dir`!):

```
project:
  type: website
  output-dir: docs
```

To create the file, you can navigate to the folder and create a new file.

- On windows, first make sure that file extensions show when you look at files in folders.
  - Now right click in the folder area, select 'New' –> 'Text Document'. Change the file name and file extension (the `.txt` part) to `_quarto.yml`. Now you can open the file in RStudio or a text editor and add the text.
- On a mac, you should be able to create a file using TextEdit in your `R project` folder. Save it as a `.doc` or whatever. Then you should be able to rename the file as `_quarto.yml`. Now you can open the file in `RStudio` or a text editor and add the text.

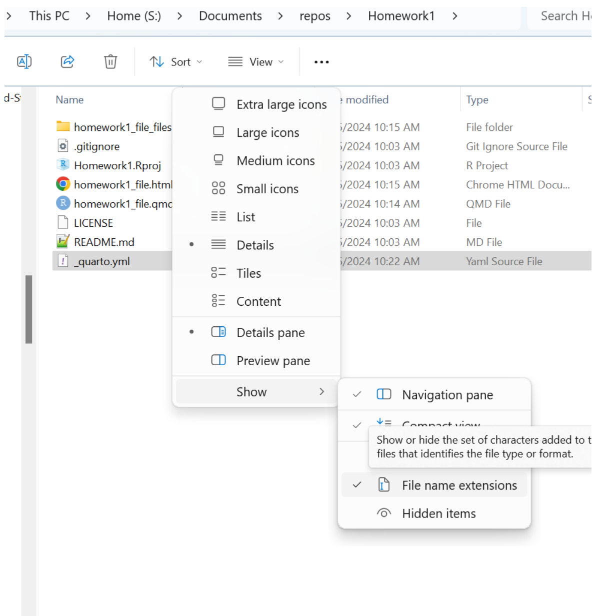

Figure 2: Show File Name Extensions

Step 5

Now you can use a similar process to create a file called `.nojekyll` in the project repo on your computer. This file doesn't need to have anything in it! You just need that file there. Note: On Windows, once you make this file, it may not show (files that start with a period are 'hidden' files). Don't worry, `git` still tracks this file. You can also go to the same 'View' menu as before and select 'Hidden items' to show these types of files.

Step 6

Now head back to `RStudio`. Open the terminal in `RStudio` (near the console there should be a few tabs, choose the Terminal). In that terminal, run the following code:

```
quarto render
```

Step 7

We did everything we need. We've set up the `.nojekyll` file to keep `GitHub` from render the main repo and we've created the `_quarto.yml` file to output our quarto files in a `doc` folder. Now we want to push all changes up to your repo on `GitHub`! You can do this via menus in `RStudio` or via the command line. If you can't get either of those to work, you can use the `GitHub` web interface for now.

Step 8

Now we just need to change a few things on `GitHub`. Head to your `GitHub` repo page. Go to settings, choose pages, and under "Branch" choose 'main' and change the folder to '/docs'. Then hit save!

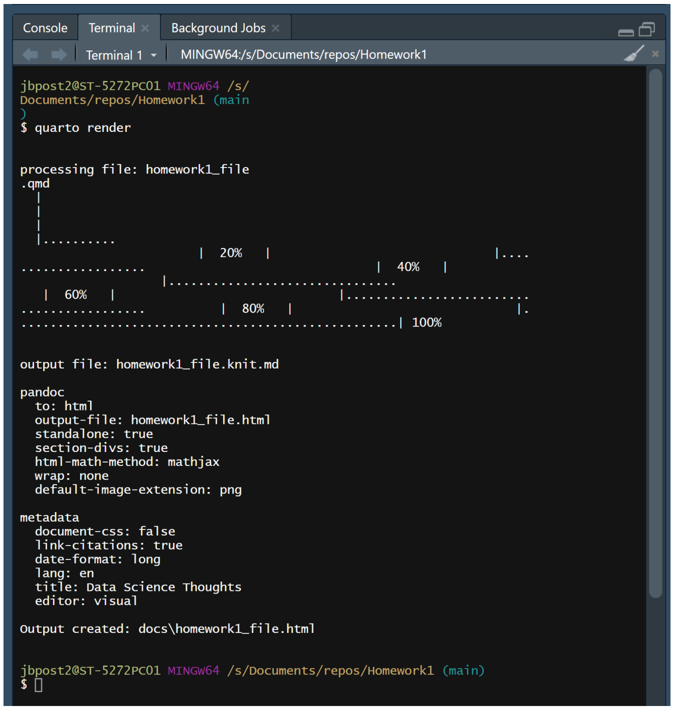

Figure 3: Image of the RStudio terminal after running quarto render

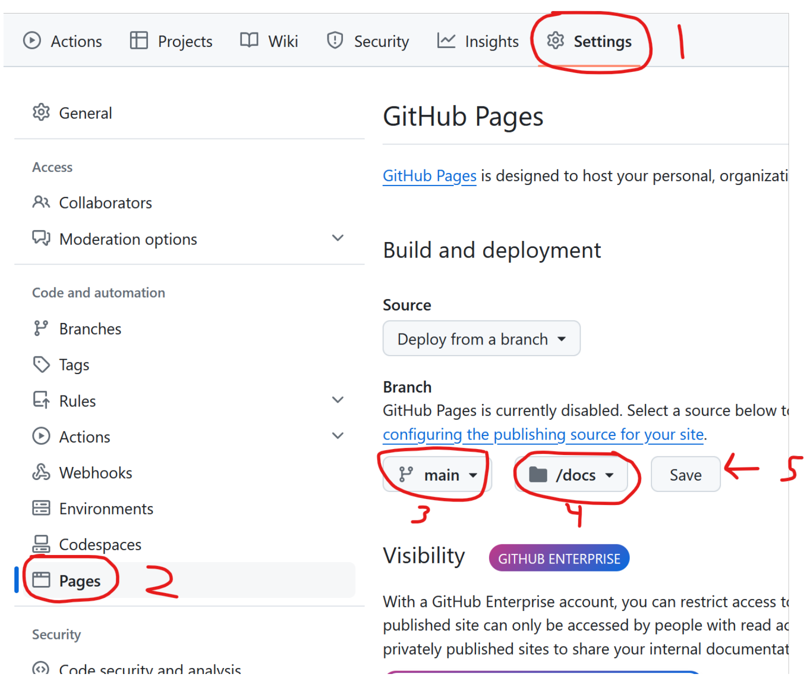

Figure 4: Options to choose in the Settings of your GitHub repo.

Step 9

Here we have to be patient! Wait about 2 minutes then head back to your main **GitHub** repo page. There will now be a 'Deployments' section on the bottom right of the page.

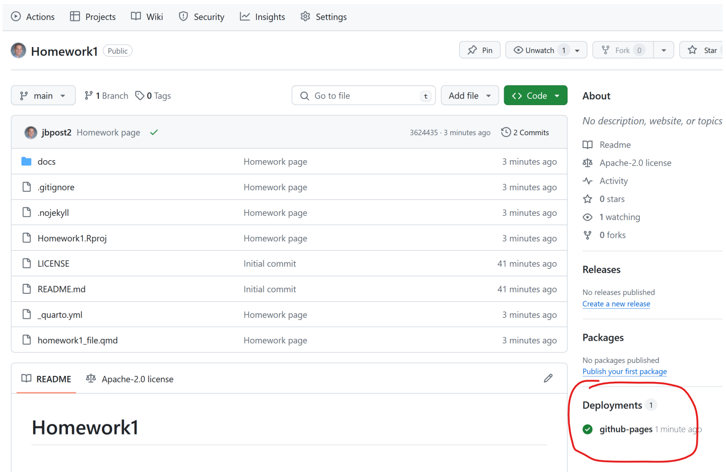

Figure 5: Deployment section shown.

Click on that. Deployments of your 'rendered' website are listed here. A green check mark appears next to the deployment when it is ready! The link below that is the link to the 'rendered' site.

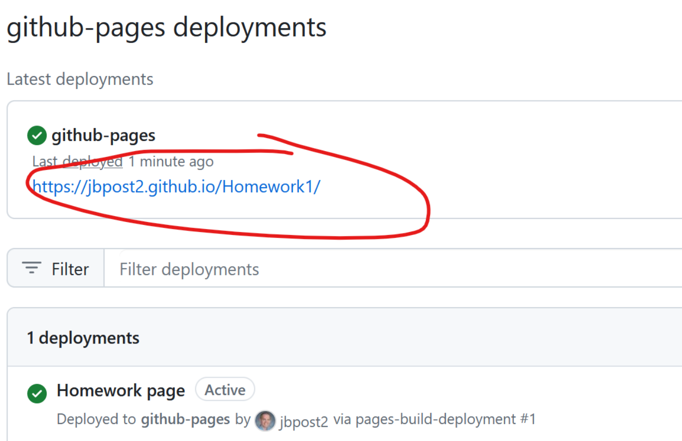

Figure 6: Link to website.

Copy the link to that site and turn this in!